

CHES 2026 TUTORIAL · ANTALYA, TÜRKİYE

Quantum Physical Unclonable Functions: Taxonomy, Implementations, and Open Problems: Lab Session

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Getting the environment ready

- 01 Clone the repo** (notebooks + Pictures/ folder)

```
git clone https://github.com/oguznrl/QPUF-Tutorial
```

- 02 Create a virtual environment** with Python 3.10–3.12 (Windows: qpuf-env\Scripts\Activate.ps1)

```
python3 -m venv qpuf-env && source qpuf-env/bin/activate
```

- 03 Install everything listed in requirements.txt** (pypuf, TensorFlow, Qiskit, Jupyter)

```
pip install -r requirements.txt
```

- 04 Open the notebooks** in Jupyter or VS Code and select the **qpuf-env** kernel. On Google Colab, skip the venv and just run the pip cells.

- 05 TypeError: Expected int8?** Run the patch cell after the attack, then re-run the attack.



Scan for the repo

`github.com/ oguznrl/
QPUF-Tutorial`

Cloning a classical PUF in software

Physically unclonable, but mathematically modelable.

01

Goal

- Simulate a 64-bit, $k=5$ XOR-APUF with pypuf
- Measure quality metrics: reliability, uniqueness, uniformity
- Run an LRAttack2021 modeling attack on 200,000 CRPs

02

What we want to see

- Additive delay model: each component is linear in $\varphi(c)$
- XOR breaks linearity, but does not remove learnability
- Good quality metrics do not mean resistance to ML attacks

03

Expected result

- Uniqueness ≈ 1 , uniformity ≈ 0.5
- Reliability < 1 : noise accumulates through the XOR
- High attack accuracy: **$> \sim 90\%$** $\rightarrow$ a working software clone

Hiding the response inside a quantum state

How much harder does the same attack get when the response arrives as a qubit?

01

Goal

- Encode the same CPUF's response from Lab 1 into a qubit via BB84
- Derive the basis bit from a second, independent PUF
- Run the same LRAttack2021 against a single-copy Eve

02

What we want to see

- No-cloning and non-orthogonal states: Eve cannot learn the basis
- Even the optimal measurement (Breidbart, Helstrom bound) leaves ~14.6% label noise
- The analytical shortcut and the Qiskit circuit give the same result

03

Expected result

- $p_{\text{guess}} \approx 0.8536$, $p_{\text{error}} \approx 0.1464$ with both methods
- No accuracy ceiling: required CRPs $\approx 1/(1-2p)^2 = 2\times$
- Attack accuracy **~50%**, vs. **>~90%** in Lab 1